

Technology #13

Self-Referencing Piezoresistive Pressure
Sensor

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Self-Referencing Piezoresistive Pressure Sensor

WHY DOES IT MATTER?

Chronic ICP monitoring sensors suffer from drift

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 4 governing equations, 11 design inputs, 7 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Piezoresistive Sensing + Tubing Mechanics + Drift Compensation. 7-step build plan with specific test articles and acceptance criteria. Target buyer: Implantable sensor company; MEMS manufacturer.

WHAT DOES THE BUYER GET?

Self-referencing sensor concept + R1 correction (drift compensation architecture); Piezoresistive + Wheatstone bridge governing equations; Bench test protocols (calibration, drift, hermeticity, kink, EMC); External precedent catalog (MEMS pressure sensors, drift compensation); Critical UNKNOWN disclosure (achievable drift, hermeticity, sensor survival); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production MEMS sensor design (diaphragm, piezoresistors, geometry); Self-referencing bridge architecture + signal conditioning; Hermetic packaging process; Catheter integration manufacturing; Regulatory submission (active implantable, ISO 14708-1, IEC 60601-1-2); Clinical validation (accuracy + drift over implantation duration); Long-term reliability testing

WHAT IS THE NEXT DECISIVE EXPERIMENT?

MEMS sensor prototypes (various diaphragm thicknesses)

WHAT WOULD MAKE US KILL IT?

Self-referencing cannot reduce drift below 1 mmHg/month (insufficient for clinical utility)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: MEMS sensor prototypes (various diaphragm thicknesses)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - MEMS sensor prototypes (various diaphragm thicknesses) - 16 weeks (foundry cycle)

Basis: Build plan WP. Scope: Sensitivity, linearity, hysteresis. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Packaged sensor (parylene or Ti canister) - 8 weeks

Basis: Build plan WP. Scope: Hermeticity per MIL-STD-883. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Chronic aging specimens - 24-52 weeks (chronic)

Basis: Build plan WP. Scope: Drift rate over time. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 4 equations. Key assumptions: Piezoresistive coefficients stable over implantation lifetime (UNKNOWN for chronic); Bridge elements matched to within tolerance; Diaphragm deformation linear with pressure; Self-referencing architecture cancels thermal drift (UNKNOWN — depends on matching); Sensor packaged to survive catheter environment (UNKNOWN). UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: MEMS sensor prototypes (various diaphragm thicknesses). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Piezoresistive MEMS sensor with self-referencing Wheatstone bridge for drift compensation (R1 fix)

POTENTIAL VALUE DRIVER: Addresses: Chronic ICP monitoring sensors suffer from drift

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Implantable sensor company; MEMS manufacturer

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Piezoresistive Sensing + Tubing Mechanics + Drift Compensation

Disciplines: MEMS engineering (piezoresistive sensors), Tubing mechanics (kink, fatigue, patency), Circuit design (Wheatstone bridge, signal conditioning), Calibration and drift compensation

Architecture: Catheter-integrated piezoresistive pressure sensor with self-referencing architecture for drift compensation. R1 fix: original P-27 suffered from chronic drift; R1 redesign adds self-referencing bridge for drift correction.

Subsystems:

SS-01: Piezoresistive sensing element (MEMS) - Converts mechanical stress to resistance change via piezoresistive effect

SS-02: Self-referencing bridge (R1 fix) - Compensates for thermal and temporal drift by referencing active sensor to dummy/passive element

SS-03: Catheter integration (mechanical protection) - Mounts sensor in catheter wall; protects from mechanical damage while transmitting pressure

SS-04: Signal conditioning + telemetry - Amplifies, filters, digitizes sensor signal; transmits to external reader

2. Mechanism Architecture

Physical Changes: Pressure on catheter deforms MEMS diaphragm; piezoresistive elements on diaphragm change resistance; Wheatstone bridge outputs voltage proportional to pressure

Key Physics: Piezoresistive effect: $dR/R = \pi_i \cdot \sigma$, where π_i is piezoresistive coefficient (typically 10^{-11} to 10^{-10} Pa $^{-1}$ for silicon, much higher than metal strain gauges). Wheatstone bridge: $V_{out} = (dR / (4R + 2 \cdot dR)) \cdot V_{ex} \approx (dR / (4R)) \cdot V_{ex}$ for small dR .

3. Governing Engineering Model

Summary: Piezoresistive pressure sensing with Wheatstone bridge and self-referencing drift compensation. Bridge output proportional to pressure-induced resistance change; dummy element cancels thermal drift.

Equations:

$dR/R = \pi_{iL} \cdot \sigma_L + \pi_{iT} \cdot \sigma_T$ [piezoresistive effect]

$V_{out} = (dR / (4R + 2 \cdot dR)) \cdot V_{ex}$ [bridge output]

$V_{out} = ((R_{active} - R_{dummy}) / (4R)) \cdot V_{ex}$ [self-referencing bridge]

$P_{measured} = V_{out} / (S \cdot V_{ex})$ [pressure from voltage, S = sensitivity]

Assumptions:

- Piezoresistive coefficients stable over implantation lifetime (UNKNOWN for chronic)
- Bridge elements matched to within tolerance
- Diaphragm deformation linear with pressure
- Self-referencing architecture cancels thermal drift (UNKNOWN — depends on matching)
- Sensor packaged to survive catheter environment (UNKNOWN)

Boundary Conditions:

- Pressure range: 0-40 mmHg (ICP range)
- Temperature: 37°C constant (CSF environment)
- Bridge excitation: typically 1-5 V
- Sensor size: constrained by catheter geometry

Failure Regimes:

- Drift exceeds self-referencing capability — sensor output unreliable

- Mechanical damage to diaphragm — sensor fails
- Bridge element mismatch — thermal drift not cancelled
- Packaging failure — CSF ingress damages electronics
- Catheter kink damages sensor — mechanical failure

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Chronic ICP monitoring enables early detection of shunt failure; current standard requires lumbar puncture or external transducer	VERIFIED
DI-002	Functional requirement	Catheter-integrated piezoresistive pressure sensor with self-referencing drift compensation (R1 fix)	MODELLED
DI-003	Pressure range	EXTERNAL_PRECEDENT (0-40 mmHg ICP range)	EXTERNAL_PRECEDENT
DI-004	Piezoresistive coefficient	EXTERNAL_PRECEDENT (Si: 10^{-11} to 10^{-10} Pa $^{-1}$)	EXTERNAL_PRECEDENT
DI-005	Drift target	MODELLED (< 0.5 mmHg/month via self-referencing, R1 fix)	MODELLED
DI-006	Sensor size constraint	Constrained by catheter diameter ~2 mm	VERIFIED
DI-007	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-008	Sterilization	UNKNOWN — MEMS may be gamma-sensitive; aseptic or EtO	UNKNOWN
DI-009	EMC	UNKNOWN — required IEC 60601-1-2 if active telemetry	UNKNOWN
DI-010	Hermeticity (MIL-STD-883)	UNKNOWN — required for chronic implant	UNKNOWN
DI-011	Battery life (if active)	UNKNOWN	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	MEMS sensor design (diaphragm, piezoresistors, geometry)	ABSENT	Diaphragm thickness; Piezoresistor geometry; Process selection
DO-002	Self-referencing bridge architecture	CONCEPTUAL	
DO-003	Catheter integration design	ABSENT	Sensor mounting; Mechanical protection; Catheter geometry
DO-004	Signal conditioning + telemetry circuit	CONCEPTUAL	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Piezoresistive coefficient	EXTERNAL_PRECEDENT (Si: 10^{-11} to 10^{-10})	Pa^{-1}	Published MEMS sensor literature	
Bridge excitation V_{ex}	UNKNOWN — design choice (likely 1-5 V)	V	Power budget vs SNR tradeoff	Bench measurement to confirm proposed range
Bridge element matching	(UNKNOWN R_{process}) dependent	$\%$	MEMS process capability	
Diaphragm thickness	UNKNOWN — design choice	μm	Sensitivity vs pressure range tradeoff	
Sensor size	UNKNOWN — constrained by catheter geometry	mm^2	Catheter integration constraint	
Drift rate (target)	MODELLED ($< 0.5 \text{ mmHg/month}$)	mmHg/month	via long-term reference R_{g} design target	
Sensor accuracy	UNKNOWN — clinical target $\sim 1\text{-}2 \text{ mmHg}$	mmHg	Clinical measurement requirement	Clinical study to confirm accuracy requirement

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Chronic drift exceeds self-generating mechanisms	Self-generating mechanisms	Periodic recalibration protocol	Chronic aging study with periodic calibration	UNKNOWN achievable drift rate
Diaphragm mechanical failure	Pressure overload or mechanical stress	Overpressure clamping, diaphragm stress margin	Overpressure test, mechanical stress test	UNKNOWN in vivo mechanical environment
Bridge element mismatch	R_{active} and R_{dummy} have different process temperature	Efficient temperature compensation	Temperature cycle test	UNKNOWN achievable matching
Packaging failure (CS Hermetic)	CS Hermetic encapsulation fails	CSF damage, hermeticity test	Hermeticity Test per MIL-STD-883C	UNKNOWN long-term hermeticity
Catheter kink damage	Catheter bending exceeds sensor mechanical limit	Sensor mechanical limit	Kink test per ISO 15550	UNKNOWN survival in vivo
EMC interference	External EM fields affect sensor	Shielding + filtering + IEC 60601-1-2 compliance	EMC test per IEC 60601-1-2	UNKNOWN specific interferences
Battery depletion (if active display)	Battery exhausted before target life	Low-power design + duty cycle management	Battery life test, energy harvesting (PNE)	UNKNOWN achievable battery life

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Chronic drift exceeds self-calibrating drift not cancelled	Temperature drift not cancelled	Periodic recalibration + spare capacity	UNKNOWN achievable drift rate
Diaphragm mechanical damage	Overpressure or impact	Overpressure protection + margin	UNKNOWN in vivo mechanical environment
Bridge element mismatch	R_active vs R_dummy temperature coefficient mismatch	Finite element control + laser trimming	UNKNOWN achievable matching
Packaging failure (CSF ingress)	Hermetic encapsulation fails	Hermetic packaging + leak test	UNKNOWN long-term hermeticity
Catheter kink damages sensor	Bending exceeds sensor limit	Sensor placement + mechanical protection	UNKNOWN sensor survival
EMC interference	External EM fields affect signal	Shielding + filtering + IEC 60601-1-2	UNKNOWN specific interferences
Battery depletion (if active)	Battery exhausted	Low-power design + duty cycling + energy harvesting	UNKNOWN achievable life

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Sensor accuracy within target	Calibration vs reference	~1-2 mmHg	NOT_TESTED
V-002	Drift < 0.5 mmHg/month	Chronic aging with calibration	< 0.5 mmHg/month	NOT_TESTED
V-003	Hermeticity MIL-STD-883	Helium leak test	< 10 ⁻⁸ atm-cc/s	NOT_TESTED
V-004	Kink resistance ISO 10555-1	ISO 10555-1	No damage at bend radius	NOT_TESTED
V-005	Biocompatibility ISO 10993	ISO 10993	Pass	NOT_TESTED
V-006	EMC IEC 60601-1-2	EMC test	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	In vivo accuracy	Animal study vs reference	Clinically acceptable	NOT_PERFORMED
VAL-002	Clinical measurement accuracy	Clinical study vs standard	Non-inferior	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
MEMS sensor die	Silicon (single-crystal Si wafer)	Standard MEMS material	Process + packaging + biocompatibility + assembly standard	CANDIDATE
Hermetic packaging	Parylene-C	Standard implantable polymer coating	Hermeticity + ISO 10993 + adhesion	CANDIDATE — preferred for thin coating
Hermetic packaging	Titanium canister	Standard implantable hermetic packaging	Hermeticity + ISO 10993	CANDIDATE — preferred for long-term
Catheter body	Silicone elastomer	Standard CSF shunt material	ISO 10993 + sensor integration	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	MEMS sensor die (custom)	1	CUSTOM_COMPONENT	Silicon (Si) wafer	UNKNOWN — MEMS foundry	CRITICAL	Calibration + drift + hermeticity
02	Signal conditioning IC	1	COTS_CANDIDATE	IC (COTS)	Multiple (TI, ADI, Maxim)	Medium	Circuit simulation + bench
03	Wireless transmitter (if active)	1	COTS_CANDIDATE	IC (COTS)	Multiple	HIGH	Range + EMC
04	Hermetic packaging (parylene-C or titanium canister)	1	CUSTOM_COMPONENT	Parylene-C or titanium	Multiple	CRITICAL	Hermeticity + biocompatibility
05	Catheter body with sensor mount	1	CUSTOM_COMPONENT	UNKNOWN — silicone elastomer	UNKNOWN — catheter manufacturer	CRITICAL	Mechanical + biocompatibility

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no catheter-integrated MEMS sensor with self-referencing process validated

Candidate Processes:

- MEMS fabrication (photolithography, etching, doping): Standard MEMS foundry process - Requires MEMS foundry partnership
- Wafer dicing + die attach: Standard semiconductor assembly - Custom for catheter integration
- Parylene-C coating (or Ti canister sealing): Standard implantable packaging - Hermeticity test required
- Catheter extrusion + sensor integration: Standard catheter manufacturing - Custom for sensor mounting

14. External Evidence

Source 1: Chronically Implanted Pressure Sensors: Challenges and State of

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4279503>

Excerpt: Drift and Long-Term Accuracy. To achieve reliable long-term measurements, implanted sensors should possess a stable, consistent response over their lifetime.

Source 2: A simple and effective method to compensate the thermal drift of

URL: <https://www.sciencedirect.com/science/article/abs/pii/S0924424724006095>

Excerpt: This article presents a study of implantable blood pressure sensors based on a probe made up of four piezoresistors. An original method combining both ...

Source 3: Compensation of Pressure Sensor Drifts - Encyclopedia.pub

URL: <https://encyclopedia.pub/entry/52473>

Excerpt: Dec 8, 2023 · To compensate induced drift in measurements, several neural networks were devised and be applied to stresses caused by two thermal cycles: 260 C ...

Source 4: Implantable Blood Pressure Sensors with Analogic Thermal Drift - MDPI

URL: <https://www.mdpi.com/2673-4591/6/1/34>

Excerpt: May 17, 2021 · To quantify and compensate for the thermal drift, we developed the equivalent model of the piezoresistive probe by using the Cadence software.

Source 5: [PDF] piezoresistive pressure sensor with dual-unit configuration for

URL: https://engineering.purdue.edu/oxidemems/conferences/transducers2013/PDFs/Papers/444_0996.pdf

Excerpt: For the first time, we propose an on-chip dual-unit sensor configuration in this study to eliminate temperature-drift and packaging-stress of pressure sensors.

15. Transfer Boundary

YOU RECEIVE:

- + Self-referencing sensor concept + R1 correction (drift compensation architecture)
- + Piezoresistive + Wheatstone bridge governing equations
- + Bench test protocols (calibration, drift, hermeticity, kink, EMC)
- + External precedent catalog (MEMS pressure sensors, drift compensation)
- + Critical UNKNOWN disclosure (achievable drift, hermeticity, sensor survival)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production MEMS sensor design (diaphragm, piezoresistors, geometry)
- > Self-referencing bridge architecture + signal conditioning
- > Hermetic packaging process
- > Catheter integration manufacturing
- > Regulatory submission (active implantable, ISO 14708-1, IEC 60601-1-2)
- > Clinical validation (accuracy + drift over implantation duration)
- > Long-term reliability testing

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable drift rate with self-referencing — UNKNOWN (critical for clinical utility)
- MEMS process capability for bridge matching — UNKNOWN
- Long-term hermeticity in CSF environment — UNKNOWN
- Sensor survival in catheter mechanical environment — UNKNOWN
- Power budget for active telemetry — UNKNOWN
- Regulatory pathway (active implantable, ISO 14708-1) — UNKNOWN specifics
- Clinical accuracy requirement — UNKNOWN

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	MEMS sensor prototype (foundry process)	Pressure chamber, bench	Sensitivity, drift	Sensitivity meets target, drift < 10 ⁻⁸ atm/yr	MEMS sensor spec	16 weeks (foundry cycle)
WP-02	Self-referencing bridge circuit	Bench circuit, temperature chamber	Bridge matching, drift compensation	Bridge matching < 10%, drift compensation < 10%	Bridge circuit spec	10 weeks
WP-03	Packaged sensor (partial hermeticity test)	Hermeticity test equipment	Hermeticity per MIL-STD-883C	Hermeticity < 10 ⁻⁸ atm/yr	Packaging spec	8 weeks
WP-04	Catheter-integrated sensor	Mechanical test frame, kink test rig	Kink resistance, drift	Kink resistance < 10 ⁻⁸ atm/yr, drift < 10 ⁻⁸ atm/yr	Catheter integration spec	10 weeks
WP-05	Chronic aging specimen	Aging chamber at 37°C	Drift CSF over time	Drift < 10 ⁻⁸ atm/yr	Drift report	24-52 weeks (chronic)

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-06	Biocompatibility specifications (MEMS sensor)	ISO 10993	ISO 10993	ISO 10993	ISO 10993 report	12 weeks (external)
WP-07	EMC test articles (MEMS sensor + signal conditioner)	EMC test articles	EMC test articles	EMC test articles	EMC report	8 weeks (external)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 4 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: Implantable sensor company; MEMS manufacturer.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Self-referencing cannot reduce drift below 1 mmHg/month (insufficient for clinical utility)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Self-referencing cannot reduce drift below 1 mmHg/month (insufficient for clinical utility)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.